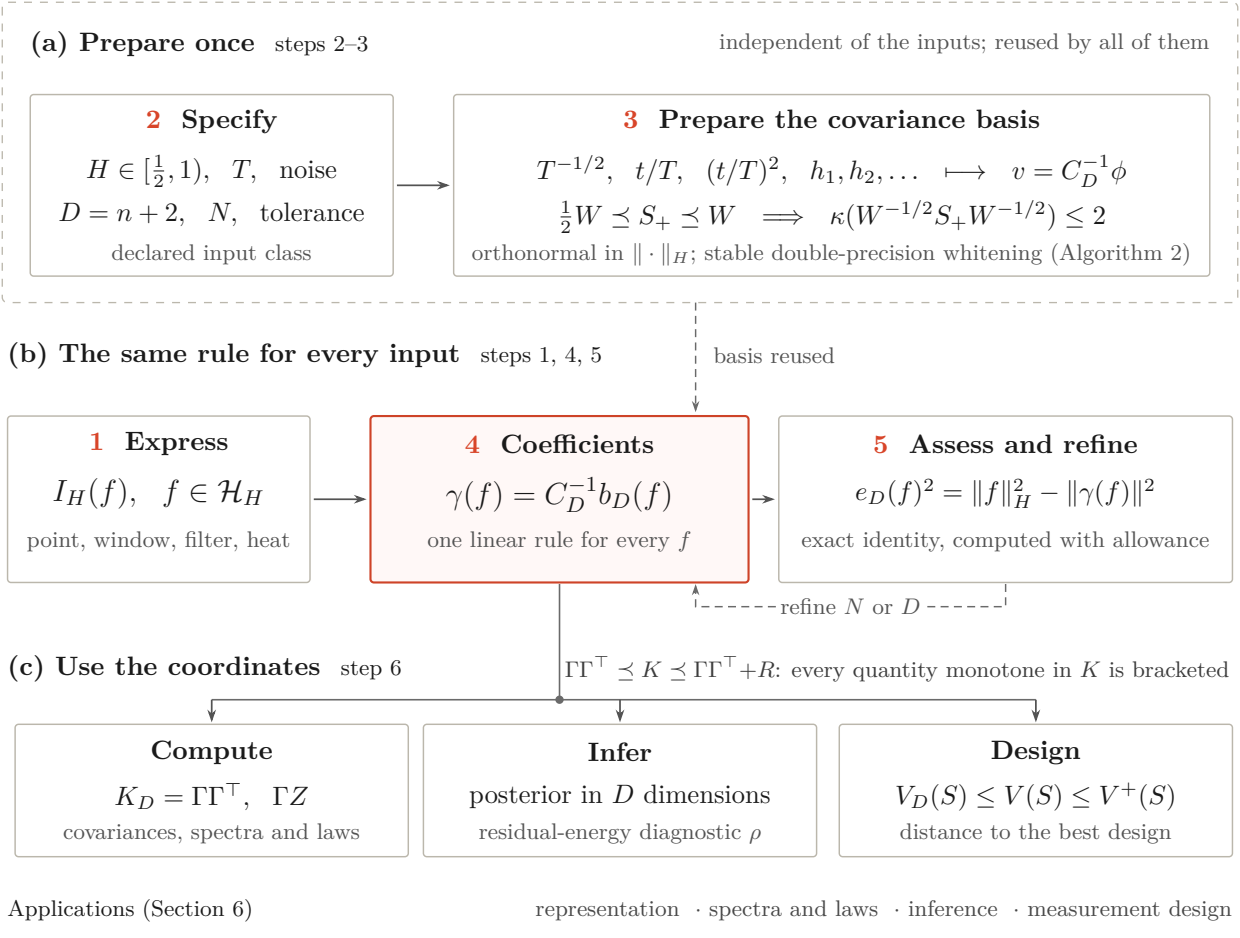

(a) Prepare once steps 2–3
independent of the inputs; reused by all of them
2 Specify
$H \in [\frac{1}{2}, 1)$, T, noise
$D = n + 2$, N, tolerance
declared input class
3 Prepare the covariance basis
$T^{-1/2}$, t/T, $(t/T)^2$, $h_1, h_2, \ldots \longmapsto v = C_D^{-1}\phi$
$\frac{1}{2}W \preceq S_+ \preceq W \implies \kappa(W^{-1/2} S_+ W^{-1/2}) \leq 2$
orthonormal in $\|\cdot\|_H$; stable double-precision whitening (Algorithm 2)
(b) The same rule for every input steps 1, 4, 5
basis reused
1 Express
$I_H(f)$, $f \in \mathcal{H}_H$
point, window, filter, heat
4 Coefficients
$\gamma(f) = C_D^{-1} b_D(f)$
one linear rule for every f
5 Assess and refine
$e_D(f)^2 = \|f\|_H^2 - \|\gamma(f)\|^2$
exact identity, computed with allowance
refine N or D
(c) Use the coordinates step 6
$\Gamma\Gamma^\top \preceq K \preceq \Gamma\Gamma^\top + R$: every quantity monotone in K is bracketed
Compute
$K_D = \Gamma\Gamma^\top$, ΓZ
covariances, spectra and laws
Infer
posterior in D dimensions
residual-energy diagnostic ρ
Design
$V_D(S) \leq V(S) \leq V^+(S)$
distance to the best design
Applications (Section 6)
representation · spectra and laws · inference · measurement design